BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Semester Final | Semester: Summer 2026
Duration: 2 hours | Full Marks: 50

CSE 320/EEE361/ECE361: Data Communications

Answer the following questions.
Figures in the right margin indicate marks.

SET B

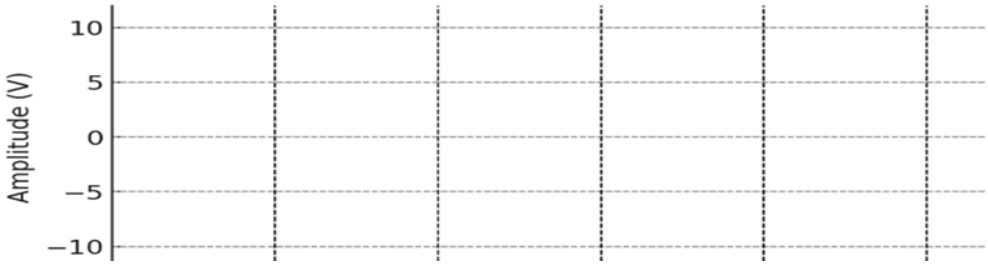
Name:	ID:	Section:

1. [CO2]

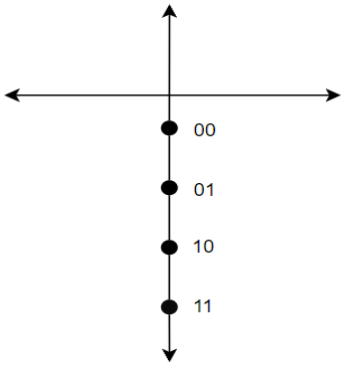
a) You have been tasked to combine two different analog signals. For the left bit signal, 0 has been phase shifted by 180^0 and 1 has been shifted by 0^0. For the right bit signal, 0 has been phase shifted by 90^0 and 1 has been shifted by 270^0. The carrier signal has two complete waves in a signal element with an amplitude of 5V. [2+3]

i. **Find** the phase shift and amplitude of the resulting signal. **Draw** the constellation diagram to show the details.

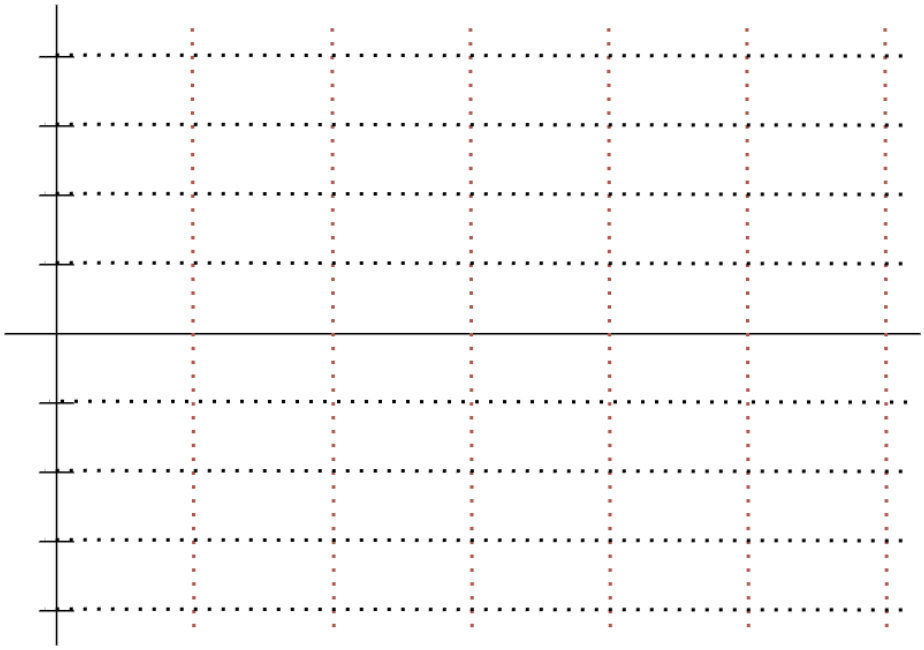
ii. **Illustrate** the analog signal for the data **1101001001**.



b) [1+4]



i. **Identify** the shift keying technique which is being depicted by the above constellation diagram. [Frequency is constant]

		ii. **Draw** the analog signal for the bit stream **0110001100** using the constellation diagram given above [frequency = 2 hz for each signal element]. You can assume the voltage levels here. 	
2. [CO3]	a)	An analog system uses FDM to multiplex 6 input channels (with guard bands) onto a higher bandwidth link. After multiplexing, the lowest frequency in the output link is 100 kHz, and the highest frequency is 800 kHz. The bandwidth of each guard band is 20% of the bandwidth in each input channel. i. **Calculate** the bandwidth in each input channel. ii. **Show** the frequency domain representation. Clearly label the frequencies for each channel.	[3+2]
	b)	Five digital channels, each with input bit rate of 40 kbps, are multiplexed using TDM. During this multiplexing, 8 interleaved bits and 3 synchronizing bits are used. The output link of this multiplexer is called a group. Four such groups are then multiplexed to create a supergroup. This second stage TDM process uses 5 interleaved bits and 3 synchronizing bits. i. **Show** the whole TDM scheme using a figure. ii. **Find** the output bit rate of a group. iii. **Calculate** output slot duration and frame rate in the supergroup.	[2+2 +3]
	c)	Five channels (A, B, C, D, E) are sharing a digital link through TDM. In the first time slot, only channels A, D and E have data; channels B and C do not have any data. In the second time slot, only channels B, D and E have data.	[3+1]

<table>
<tr><td></td><td></td><td>i. Which TDM technique should be used here? Show the multiplexing process and the frames using a figure.

ii. When should we NOT use this technique?</td><td></td></tr>
<tr><td rowspan="2">3.
[CO4]</td><td>a)</td><td>Suppose the sender generates the Codeword 100100101, where the first 6 bits are the original Dataword.

i. What is the Dividend at the sender side? What is its length?

ii. You are given two possible Divisors 1111 and 1011. Determine which one is the correct Divisor.

iii. During transmission, the Codeword has changed to 100110101. Using the correct Divisor from ii, find out and comment on whether the CRC method is capable of detecting the error on the receiver side. Show the necessary steps.</td><td>[1+3 +3]</td></tr>
<tr><td>b)</td><td>Juliette wants to send the message “HELLOWLD” to her friend Shirley using the Internet Checksum method. Assume that the message is divided into 16-bit words, with each character represented using its hexadecimal ASCII value.
The hexadecimal values of the characters are as follows:

<table><tr><th>Character</th><th>Hexadecimal Value</th></tr><tr><td>D</td><td>44</td></tr><tr><td>E</td><td>45</td></tr><tr><td>H</td><td>48</td></tr><tr><td>I</td><td>49</td></tr><tr><td>L</td><td>4C</td></tr><tr><td>N</td><td>4E</td></tr><tr><td>O</td><td>4F</td></tr><tr><td>W</td><td>57</td></tr></table>
i. Shirley receives the checksum as CFE1. Verify whether this was the actual checksum calculated from Juliette’s side. Show the calculations in detail.

ii. Along with the checksum, Shirley receives the message “IELLNWLD”. Verify whether the received data contains a detectable error. Show the calculations in detail and justify your answer.</td><td>[2+3]</td></tr>
</table>

4. **[CO4]**	a)	Four stations (W, X, Y, Z) share a single channel using the CSMA protocol. Distances between the stations and propagation speeds are mentioned below. The channel bandwidth is 20 Mbps. **W to X:** 600 m of coaxial cable (speed = 2×10^8 m/s) **X to Y:** 1500 m of fiber-optic cable (speed = 2.5×10^8 m/s) **Y to Z:** 800 m of coaxial cable (speed = 2×10^8 m/s) i. **Calculate** the maximum propagation delay between the stations and determine the vulnerable time in this scenario. ii. Station W begins transmitting a 250-bit frame at t = 0. Using the delay found in (i), **determine** the time at which the first bit of W's frame arrives at the farthest station, and the frame transmission time (T_{fr}). iii. **Determine** whether W's 250-bit frame is large enough to guarantee collision detection before transmission ends. If not, **calculate** the minimum frame size (in bits) that would satisfy this requirement. iv. Suppose the stations use non-persistent CSMA. Briefly **explain** how it can lead to unnecessary idle time in the channel. Which CSMA technique would you recommend instead?	[3+2 +3+2]
	b)	A satellite communication system uses a shared channel using the **slotted ALOHA** protocol. Both satellites use the same frequency/channel and have overlapping coverage, so their transmissions can interfere. The channel has a bandwidth of 10 Mbps, and each frame size is 5 kB (kilo-bytes). Each slot length is the same as the frame transmission time, starting from t = 0. Satellite-1 generates a frame at t=6 ms, while satellite-2 generates a frame at t=9 ms. **Explain** whether it will lead to a collision or not. *Hint: "generates a frame and transmit a frame are different."*	[2]

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